

Assignment 2

Q1

(i)

Deviance is always non-negative because the full model has fewer restrictions than the restricted so mathematically its maximized log-likelihood must be at least as large as the restricted model.

(ii)

Large deviances may indicate that the restricted model fits the data worse than the full model. So removing those predictors caused a significant loss of the fit. This suggests that those predictors may be important.

(iii)

Under H_0 the deviance is approximately distributed as chi-square with p_2 degrees of freedom.

(iv)

Since the deviance can be compared to a chi-square distribution with p_2 degrees of freedom, we can obtain a p-value. If $p\text{-value} < 0.05$ we can reject the H_0 . As such we can conclude that the removal of those predictors results in a significant loss of it. If $p\text{-value} > 0.05$, then there is no significant evidence that the restricted model fits worse than the full model.

Q2

(i)

A Poisson GLM was fitted with art as the response variable and fem, mar, phd, and ment as predictor variables. The Wald tests were used to test hypotheses $H_0: \beta_j = 0$ vs $H_1: \beta_j \neq 0$. At a 5% significance level:

Poisson Model				
	☐ Estimate	Std. Error	Z-Value	P-Value
(Intercept)	0.28377964	0.10311912	2.7519595	5.923985e-03
femWomen	-0.17662819	0.05390813	-3.2764665	1.051148e-03
marMarried	0.03395383	0.05659383	0.5999563	5.485354e-01
phd	0.01322241	0.02649107	0.4991271	6.176898e-01
ment	0.02486035	0.00198675	12.5130751	6.332441e-36

- **femWomen** was statistically significant ($z = -3.2765$ and $p\text{-value} = 0.00105$),
- **ment** was statistically significant ($z = 12.5131$ and $p\text{-value} < 0.001$)
- **marMarried** and **phd** were not statistically significant ($p\text{-value} = 0.5485$ and $p\text{-value} = 0.6177$, respectively)

(ii)

Given pearson dispersion parameter $\tau = 1.8576$, since $\tau > 1$, this indicates that the observed variability in the article counts is larger than the variance assumed with a Poisson model. A Poisson model assumes $E(Y) = \text{Var}(Y)$, this suggests presence of overdispersion, i.e. the model may be underestimating the variability in the data and produce untrustworthy standard errors.

(iii)

Poisson_model			
	☐ Estimate	Std. Error	P-Value
femWomen	-0.17662819	0.05390813	1.051148e-03
ment	0.02486035	0.00198675	6.332441e-36

Quasi_model			
	☐ Estimate	Std. Error	P-Value

femWomen	-0.17662819	0.07347299	1.641635e-02
ment	0.02486035	0.00270780	2.807739e-19

A Quasi-Poisson GLM model was fitted using the same response and predictor variables as the Poisson model. Between the models both standard errors increased under Quasi-Poisson. **femWomen** increased from 0.05390813 to 0.073472996 and **ment** from 0.00198675 to 0.002707801. Since $\tau = 1.8576 > 1$ indicating overdispersion. The quasi-poisson model accounts for extra variability allowing $\text{Var}(Y) = \tau E(Y)$ allowing the quasi-poisson to provide larger standard errors and more conservative inference.

(iv)

Poisson_model			
	☐ Estimate	Std. Error	P-Value
femWomen	-0.17662819	0.05390813	1.051148e-03
ment	0.02486035	0.00198675	6.332441e-36

Nb_model			
	☐ Estimate	Std. Error	P-Value
femWomen	-0.16660662	0.07185634	2.041634e-02
ment	0.02891321	0.00323353	3.832737e-19

A Negative Binomial regression model was fitted using the same response and predictor values as the Poisson model. The coefficient estimates were similar to the Poisson Model. Predictors **femWomen** ($\beta = -0.1666$, p-value = 0.0204) and **ment** ($\beta = 0.0289$, p-value < 0.001) remained statistically significant

The predictors **marMarried** (p-value = 0.6081) and **phd** (p-value = 0.6294) remained statistically non-significant. Thus, the conclusions regarding predictor significance remain unchanged.

Though when overdispersion is present the Negative Binomial model allows for greater variability in the count response.

(v)

From the full model **art ~ fem + marMarried + phd + ment**, StepAIC removed **phd** first then **marMarried** leaving the selected model **art ~ fem + ment**. The AIC value decreased from 3143.0 for the full model to 3139.4 for the selected model, meaning stepwise prefers the smaller model. A likelihood-ratio was used to compare the 2 nested models. The test resulted in a likelihood ratio = 0.4427 with df = 2 and p-value = 0.8014. Since p-value > 0.05, we fail to reject the null hypothesis that the reduced model fits as well as the full model. Thus, the two models don't differ significantly in fit. Smaller model is preferred.

Appendix

```
# Q2 (i) Poisson GLM

# Install pscl
install.packages("pscl")

library(pscl)

# Load the bioChemists data
data("bioChemists")

# Check the dataset
str(bioChemists)
head(bioChemists)

# Check factor/reference levels
levels(bioChemists$fem)
levels(bioChemists$mar)

# Fit the Poisson GLM

poisson_model <- glm(
  art ~ fem + mar + phd + ment,
  family = poisson(link = "log"),
  data = bioChemists
)
```

```
summary(poisson_model)

# coefficient table

coef(summary(poisson_model))
```

```
# Q2 (iii)
quasi_model <- glm(
  art ~ fem + mar + phd + ment,
  family = quasipoisson(link = "log"),
  data = bioChemists
)

summary(quasi_model)

coef(summary(quasi_model))
```

```
# Q2 (iv)
library(MASS)

nb_model <- glm.nb(
  art ~ fem + mar + phd + ment,
  data = bioChemists
)

summary(nb_model)

coef(summary(nb_model))
```

```
# Q2(v)

nb_selected <- stepAIC(
  nb_model,
```

```
    direction = "both"  
  )  
  
  # See the final selected model  
  summary(nb_selected)  
  
  # Show the selected formula  
  formula(nb_selected)  
  
  # Compare selected and full Negative Binomial models  
  anova(nb_selected, nb_model, test = "LRT")
```